A Small-Data Machine Learning Framework for Forecasting Urban Air Pollution Spikes in Jaipur

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ABSTRACT

Urban air pollution, especially $PM_{2.5}$ and PM_{10}, has emerged as a serious concern in rapidly urbanizing Indian cities such as Jaipur. Machine learning algorithms have shown promise for short-term air-quality forecasting; however, their performance is often limited in cities with sparse monitoring infrastructure and small historical datasets.
This paper presents a small-data-oriented framework that combines domain-knowledge-driven feature engineering with a sample-weighted learning strategy[24] to better capture rare high-pollution events while preserving the temporal structure of the dataset. Using 8-hourly CPCB measurements collected from 2019 to 2025, we develop a Random Forest regression model that integrates domain-specific temporal and festival-related features such as Diwali and Lohri indicators. The feature set includes temporal indicators, meteorological variables, autoregressive lag features, and rolling statistics that capture short-term pollution persistence.
The enhanced $PM_{2.5}$ model achieves a mean absolute error of approximately 9.61 μg/m^3 and an $R^2 \approx 0.70$ on the held-out 2024–2025 test set, while performance during the Diwali 2025 festival period shows degradation due to distribution shift and extreme pollution spikes. The results indicate that sample-weighted learning combined with temporal feature engineering helps capture rare high-pollution events and enables useful early-warning air-quality predictions for Indian cities.

Introduction

One of the most challenging environmental issues of the twenty-first century is urban air pollution, and this is further aggravated by the rapidly increasing rate of urbanization and industrialization in developing countries. Jaipur, being one of the most prominent cities in the state of Rajasthan with a population of over three million, experiences a wide range of activities, from industrial production to agriculture, which have a direct impact on air quality. These activities contribute to increased concentrations of fine particulate matter ($PM_{2.5}$ and PM_{10}). During cultural events such as Diwali and winter festivals like Lohri, brief but intense spikes in $PM_{2.5}$ and PM_{10} trigger very poor or severe Air Quality Index (AQI) categories and increase risks of severe respiratory issues[1,16–18].

Air quality prediction models based on data analytics use past air quality and meteorological data to predict future air quality and provide early warnings[10–13]. However, most of the existing work is focused on data-rich cities such as Mumbai, Delhi, and Bengaluru, which have dense air quality monitoring stations and rich historical data[14,15]. Jaipur, on the other hand, has a remarkably small air quality monitoring dataset, which is mostly limited to a few monitoring stations and a few years of data. The small dataset causes over-optimistic bias in the model, fails to generalize to new years, and performs poorly on small and rare events[2–5]. Thus, we identify a key research gap: developing a lightweight and interpretable machine learning framework capable of learning from a relatively small dataset (approximately 7,670 eight-hourly observations) while still capturing rare festival-related pollution spikes.

Air pollution patterns in Jaipur are not random. They show daily and weekly trends, as well as seasonal trends, including spikes during festivals such as Diwali and Lohri. However, we have very limited data for such seasonal trends. This makes Jaipur a suitable example for studying machine learning methods on small environmental datasets.

Therefore, we incorporate a sample-weighted training strategy[24] that increases the importance of rare high-pollution events during model learning. In addition, we include temporal features, autoregressive lag features derived from previous $PM_{2.5}$ and PM_{10} observations, rolling statistical features capturing recent pollution trends, and domain-knowledge features such as festival indicators, thereby creating a framework that can be applied to different cities.

Key Contributions of This Work

We build a small-data-oriented forecasting model for PM2.5 and PM10 in Jaipur, using data from multiple CPCB monitoring stations with 7,670 eight-hourly observations from 2019 to 2025. This study proposes the incorporation of domain knowledge

features, including temporal features, autoregressive lag features, rolling statistical features, and Diwali and Lohri features, and demonstrates their effectiveness in enhancing the performance of Random Forest models on a test set for 2024–2025[10–13]. We evaluate the effect of sample-weighted training[24] to better capture rare extreme pollution events ($PM_{2.5} \geq 150\ \mu g/m^3$) and analyze how this strategy influences prediction performance during high-pollution periods in Jaipur, a tier-2 Indian city[19,20].

Related Work

Research studies show that forecasting in Indian cities, particularly Tier-1 cities such as Delhi and Mumbai, attains higher accuracies given the high densities of weather monitoring infrastructure across these cities[10–15]. Models like Random Forests and gradient boosting, as well as neural networks and, more recently, deep learning architectures have been employed to identify non-linear relationships between weather variables, emission indicators, and historical pollutant levels[10–15,22,23]. Most of the studies are based on extensive monitoring of the networks. Therefore, they are not applicable in Tier-2 cities such as Jaipur due to the lack of data availability.

In recent times, there has been increasing research on the difficulties of using machine learning on small datasets and how it can be prone to overfitting[2–5]. Several approaches have been proposed to address small-sample learning problems, including synthetic data generation techniques such as SMOTE[6,7]. This paper fills the gap in the identified problem by proposing a machine learning approach that is suited to the data limitations of Jaipur. The primary aim is to combine domain knowledge-driven feature engineering, time-series-safe validation, and sample weighting to address the challenges of small datasets and rare-event forecasting in urban air quality prediction.

Comparison with Previous Studies

Previous studies on air pollution prediction in Indian cities have shown that the R^2 values are generally between 0.45 and 0.65 depending on the dataset available[10,11,20]. Since Jaipur is a Tier-2 city with limited monitoring infrastructure, achieving an R^2 value of about 0.70 demonstrates that the proposed framework can produce reasonably reliable predictions even with relatively limited historical data.

Table 1. Comparison with Previous Air Pollution Prediction Studies

Study	City	Model	R^2
Sreenivasulu and Rayalu[10]	Delhi	ML Ensemble	0.45–0.55
Kumar and Pande[11]	Indian Cities	ML Models	∼0.50
Kothari et al.[20]	Multiple Indian Cities	Random Forest	∼0.99
This Study	Jaipur	Random Forest	**∼0.70**

Overall test performance (2024–2025)

On the 2024–2025 test set, the Random Forest model showed consistent performance for both PM2.5 and PM10 after incorporating autoregressive lag features, rolling statistical features, meteorological variables, and festival indicators, which are commonly used in air-quality forecasting models[10,11].

The Random Forest model achieved an MAE of approximately 9.61 $\mu g/m^3$ and an R^2 of about 0.70 for PM2.5, while for PM10 the MAE was approximately 21.90 $\mu g/m^3$ with an R^2 of about 0.59.

These results indicate that the model captures the main pollution dynamics in the dataset while maintaining stable predictive performance under small-data conditions.

Diwali 2025 performance

In the period around Diwali 2025, $PM_{2.5}$ concentrations exceeded 300 $\mu g/m^3$, causing AQI to be around 300 to 400[16–18]. Under these severe pollution conditions, the evaluation results show that the model achieved an R^2 of approximately 0.53, with an MAE of about 19.4 $\mu g/m^3$ and a MAPE of 31% for $PM_{2.5}$. For PM_{10}, the model produced a MAPE of about 38% with a relatively low $R^2 \approx 0.21$. The model tracks the timing and approximate magnitude of the spike to distinguish between moderate, poor, and severe AQI levels[10–15]. During earlier Diwali seasons from 2019 to 2024, the R^2 for $PM_{2.5}$ generally ranges between about 0.67 and 0.86, with MAPE values typically between about 12% and 21%, indicating that the model captures the general timing and magnitude of festival-related pollution spikes[16–18].

Table 2. Benchmark comparison of regression models on the 2024–2025 test set for $PM_{2.5}$ prediction.

Model	MAE (μg/m^3)	R^2
Linear Regression	11.59	0.623
Ridge Regression	11.59	0.624
Lasso Regression	11.59	0.624
K-Nearest Neighbors ($k = 5$)	12.82	0.538
SVR (RBF Kernel)	10.78	0.660
Random Forest	9.61	0.705
Gradient Boosting	12.75	0.516

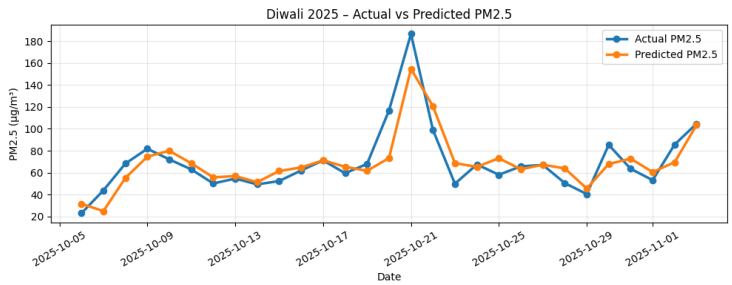

Figure 1. Observed vs. predicted $PM_{2.5}$ concentrations during the Diwali period (2025).

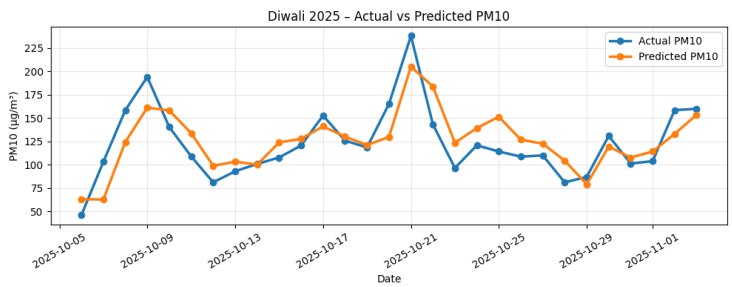

Figure 2. Observed vs. predicted PM_{10} concentrations during the Diwali period (2025).

Lohri 2025 performance

While Lohri 2025 had a very low and stable pollution level, winter conditions still contribute to slightly elevated pollution levels compared with normal periods[16–18]. During the Lohri period, PM2.5 prediction performance is evaluated primarily using MAPE because pollution levels remain relatively stable with low variance, making R^2 less reliable as a performance indicator. Across the years 2019–2023, the $PM_{2.5}$ MAPE typically ranges between approximately 16% and 19%, indicating stable predictive performance during moderate winter pollution conditions. In the later years, performance degrades slightly, with $PM_{2.5}$ MAPE values increasing to about 19.5% in 2024 and 24.2% in 2025.

For PM_{10}, the model achieves R^2 values between approximately 0.49 and 0.76 during 2019–2023, with corresponding MAPE values between about 15% and 23%. In the later years, PM_{10} performance decreases slightly, with R^2 values of approximately 0.31 in 2024 and 0.37 in 2025, and MAPE values of about 23.5% and 19.9%, respectively.

Combined with the Diwali performance results, these findings indicate that the proposed framework is able to capture general pollution patterns across both extreme pollution periods (Diwali) and relatively moderate winter conditions (Lohri), although prediction performance degrades slightly in later years due to distribution shift in Jaipur[19,20].

Distribution Shift Between 2019–2023 and 2024–2025

A significant drop in performance has been observed during the 2024–2025 test period compared to the earlier years. To understand this, we compared the statistical distribution of $PM_{2.5}$ concentrations between the training period (2019–2023) and the test period (2024–2025). The mean $PM_{2.5}$ concentration decreased from 61.72 μg/m^3 during 2019–2023 to 54.46 μg/m^3 during 2024–2025. Similarly, the variance decreased from 1363.26 to 920.03, indicating a change in the spread of pollution levels across the two periods.

These changes in mean and variance indicate the presence of distribution shift, also referred to as concept drift in machine

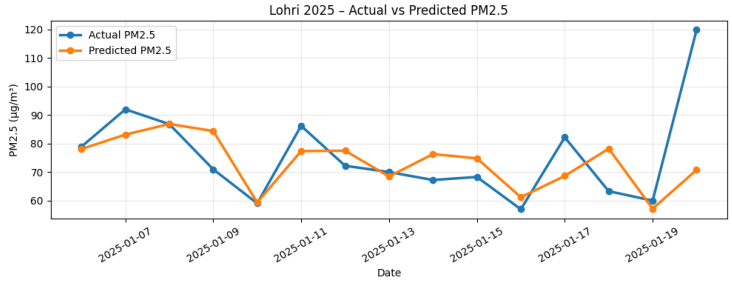

Figure 3. Observed vs. predicted $PM_{2.5}$ concentrations during the Lohri period (2025).

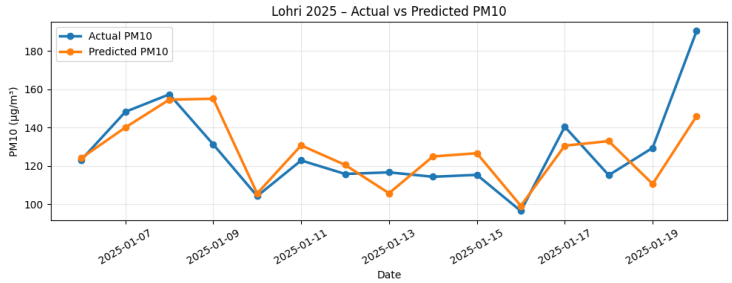

Figure 4. Observed vs. predicted PM_{10} concentrations during the Lohri period (2025).

learning[2,3]. Concept drift means that the statistical properties of the target variable change over time, which leads to models trained on older data becoming less accurate on newer data[2,3]. This shift may occur because of changes in emission sources or meteorological conditions[10–14].

Ablation: Random Forest with and without sample weighting

Applying sample weighting to high-pollution events (PM2.5 $\geq$ 150 μg/m^3 with weight 3.0) produces a modest trade-off in performance on the 2024–2025 test set. For PM2.5, R² decreases slightly from 0.705 to 0.660 while MAE increases marginally from 9.61 to 10.10 μg/m^3. PM_{10} performance remains stable (MAE 21.90 μg/m^3, R² 0.59 for both). This indicates that emphasizing rare high-pollution observations primarily improves the model's ability to capture extreme pollution spikes rather than improving overall global accuracy.

Table 3. Ablation study showing the impact of sample weighting on Random Forest performance using the 2024–2025 test set.

Model	MAE ($PM_{2.5}$)	R^2 ($PM_{2.5}$)	MAE (PM_{10})	R^2 (PM_{10})
RF without sample weighting	9.61	0.705	21.90	0.595
RF with sample weighting	10.10	0.660	21.90	0.595

Error Distribution

Fig. 6 shows the distribution of prediction errors during the Diwali and Lohri period. The residuals are approximately centered around zero, indicating that the model does not exhibit overprediction or underprediction during these periods. Larger errors occur primarily during extreme spikes, where pollution is harder to predict due to sudden increases in pollutant levels during festivals.

Note: The R^2 values for $PM_{2.5}$ during the Lohri period are not reported because the pollution levels during this period remain relatively stable with low variance, making R^2 unreliable as a performance indicator. Therefore, MAPE is used as the primary evaluation metric for $PM_{2.5}$ during Lohri. Performance remains consistent from 2019 to 2023, with R² mostly above 0.7. However, it drops in 2024 and 2025. This pattern further supports the presence of distribution shift between the training and recent test periods.

The data points on the Q–Q plot follow the reference line, which indicates the normal distribution of the residuals. Some minor deviations are observed in the Q–Q plot, which might be due to environmental factors and sudden spikes in the pollution levels.

Table 4. Year-wise model performance during the Diwali period.

Year	R^2 ($PM_{2.5}$)	MAPE (%) ($PM_{2.5}$)	R^2 (PM_{10})	MAPE (%) (PM_{10})
2019	0.870	18.50	0.704	17.90
2020	0.806	20.70	0.686	19.10
2021	0.812	16.60	0.757	14.30
2022	0.798	20.00	0.810	18.50
2023	0.699	12.00	0.733	13.20
2024	0.485	17.40	0.516	15.30
2025	0.549	27.40	0.396	28.60

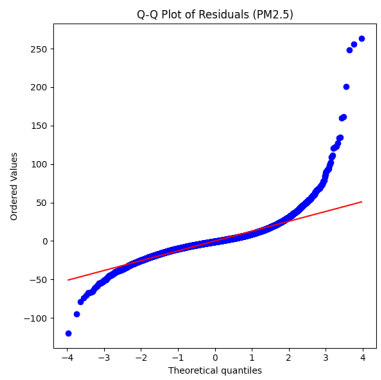

Figure 5. Q–Q plot of residual errors for $PM_{2.5}$ predictions showing approximate normality.

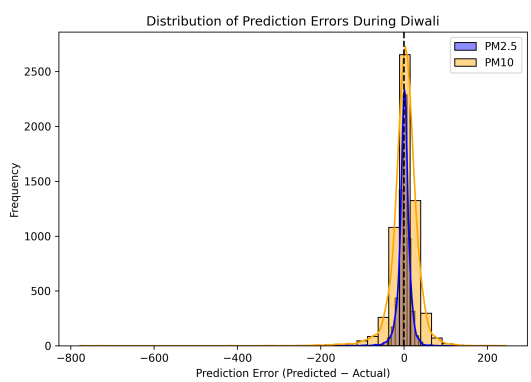

Figure 6. Histogram of prediction residuals for $PM_{2.5}$ and PM_{10} during the Diwali and Lohri periods.

Table 5. Year-wise model performance during the Lohri period (±7 day window).

Year	MAPE (%) ($PM_{2.5}$)	R^2 (PM_{10})	MAPE (%) (PM_{10})
2019	18.1	0.531	22.5
2020	16.6	0.587	17.1
2021	18.6	0.627	15.5
2022	18.7	0.760	17.7
2023	17.6	0.492	18.6
2024	19.5	0.307	23.5
2025	24.2	0.373	19.9

The residuals are normally distributed around zero, indicating that the model does not exhibit systematic overprediction or underprediction of pollution levels. This behavior suggests that the Random Forest model produces unbiased predictions across both moderate and extreme pollution conditions.

Discussion

The primary focus of this study is to evaluate how well machine learning performs when data is limited in Tier-2 cities[2–5]. The model combines pollutant concentrations with domain-knowledge features such as temporal indicators and festival-related variables to improve prediction of rare high-pollution events.

Future research may involve expanding the framework to include additional monitoring stations and applying this approach across multiple Indian cities. By incorporating local domain knowledge, such as information about local festivals, industrial patterns, or seasonal crop burning, and using the same strategy, this framework can be applied to Tier-2 and Tier-3 cities with limited monitoring infrastructure[19,20].

This study presents a framework for forecasting urban air pollution "spikes" in Jaipur using machine learning strategies suitable for small-dataset and rare-event forecasting scenarios. The enhanced Random Forest model, by combining autoregressive lag features derived from previous $PM_{2.5}$ and PM_{10} observations, rolling statistical features, meteorological variables, and appropriate festival indicators for Diwali and Lohri, achieved reasonably strong predictive performance on the overall test set for both $PM_{2.5}$ and PM_{10}, indicating its potential for providing early-warning support during critical pollution periods[10,11].

Methods

Data collection and preprocessing

Air quality data were taken from CPCB monitoring stations in Jaipur. This data spans from January 2019 to December 2025 and is recorded every 8 hours. The data includes concentrations of $PM_{2.5}$, PM_{10}, NO, NO_2, NH_3, SO_2, CO, and ozone. Raw files from each year were merged and column names were made consistent. Negative pollutant values were removed, which were considered data-entry errors. Missing entries, about 0.4 to 1.0% per column, were filled using forward-fill and linear interpolation to maintain the timeline. After cleaning, there were 7,670 valid observations left.

Feature engineering

We developed four types of features:

Temporal Features: Features were extracted from timestamps, including hour of day, day of week, and month. To preserve the cyclical nature of daily pollution patterns, the hour feature was encoded using sine and cosine transformations.[10–13].

Domain-knowledge features: We used binary indicators and continuous variables for major festivals. For Diwali, we created a binary flag for all dates within a window of ±21 days of the Diwali date each year, and added a continuous variable *days_since_diwali* to calculate the distance from the Diwali date. Similarly, for Lohri, dates within ±7 days of January 13 each year were considered[16–18].

Meteorological Features: Meteorological variables such as temperature, humidity, and wind speed were included as predictors because they influence pollutant dispersion and atmospheric conditions.

Autoregressive features: Past pollution values were used to predict current concentrations. For $PM_{2.5}$, we included lagged values at 1, 2, 3, and 24 time steps ($PM_{2.5,t-1}$, $PM_{2.5,t-2}$, $PM_{2.5,t-3}$, $PM_{2.5,t-24}$), capturing short- and medium-range persistence. For PM_{10}, we included lag-1 and lag-2 terms ($PM_{10,t-1}$, $PM_{10,t-2}$). In addition, a 24-step rolling mean and rolling standard deviation of $PM_{2.5}$ were used to summarize recent local trends and variability[10–13,22,23].

Model development

We trained separate Random Forest regression models for the prediction of $PM_{2.5}$ and PM_{10} under two configurations :

Baseline model: It is based on temporal, meteorological, and lagged pollutant features. The training data includes observations from 2019 to 2023 (about 5,247 samples), while the test set includes data from 2024 to 2025 (around 2,423 samples).

Enhanced model: In order to enhance the prediction of rare high-pollution events, we applied a sample-weighted learning strategy[24]. Observations with $PM_{2.5}$ concentrations greater than or equal to 150 µg/m^3 were assigned higher training weights, while normal observations retained standard weights. This approach emphasizes rare extreme pollution events during model learning while preserving the chronological structure of the time-series data.

The Random Forest was selected due to its ability to handle nonlinear relationships, robustness to multicollinearity, and strong performance on small-to-medium datasets. Moreover, unlike deep learning models, Random Forest is less susceptible to overfitting when the training data is small.

In addition to the fixed train–test split (2019–2023 training and 2024–2025 testing), we conducted a rolling yearly evaluation to assess temporal generalization. The rolling process involved training and testing the models using a cumulative dataset up to a particular year. For instance, if the model was trained using data from 2019 to 2021, its accuracy will be tested using data from 2022. This cycle is continued for each year.

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} f_t(x) \tag{1}$$

where T is the number of trees in the forest and $f_t(x)$ represents the prediction of the t-th decision tree.

Statistical Validation

We performed TimeSeriesSplit (5 folds) cross-validation on the 2019–2023 training data to measure model stability while respecting temporal order. Each fold trains on all data up to a split point and tests only on the subsequent temporal window. TimeSeriesSplit ensures no future data leaks into training, addressing a key methodological concern for time-series forecasting[21–23]. Cross-validation scores for $PM_{2.5}$: [0.635, 0.562, 0.499, 0.591, 0.712], mean $R^2 = 0.600$, std = 0.071.

To evaluate whether the enhanced model provides statistically significant improvements, we conducted paired statistical tests on the absolute prediction errors of the baseline and enhanced models using the 2024–2025 test dataset. A paired t-test and a Wilcoxon signed-rank test were applied separately for the $PM_{2.5}$ and PM_{10} predictions.

Table 6. Statistical significance of model predictions compared to a zero-error baseline.

Pollutant	Paired t-test p-value	Wilcoxon p-value
$PM_{2.5}$	0.629	2.53×10^{-20}
PM_{10}	0.629	2.53×10^{-20}

The paired t-test indicates that the mean prediction error is not significantly different from zero ($p \approx 0.63$), while the Wilcoxon signed-rank test detects a very small but statistically significant median deviation due to the large sample size. However, the magnitude of this deviation is negligible compared with the large pollution spikes observed during Diwali and Lohri.

Model Configuration

We benchmarked seven regression models on identical train/test splits using the complete time-series feature set (autoregressive lag features ($t-1, t-2, t-3, t-24$ for $PM_{2.5}$; $t-1, t-2$ for PM_{10}), rolling windows, festival indicators, and meteorological variables):

- Linear Regression
- Ridge Regression ($\lambda = 1.0$)
- Lasso Regression ($\lambda = 0.001$)
- K-Nearest Neighbors ($k = 5$)
- SVR (RBF kernel, $C = 10$, $\varepsilon = 0.1$)
- Random Forest (n_estimators = 150, max_depth = 10)

- Gradient Boosting

All models used scikit-learn with identical train/test splits (2019–2023 training, 2024–2025 testing). Random Forest hyperparameters: n_estimators = 150, max_depth = 10, random_state = 42. Data from 2019–2023 were used for training, while data from 2024–2025 were reserved for testing. We controlled model complexity by limiting tree depth, which helped reduce overfitting, especially under small data conditions.

Author Contributions

Riddhi Sharma: Writing – original draft, Data curation, Visualization, Methodology, Investigation. Aditya Kumar Jha: Software implementation, Validation, Formal analysis, Visualization, Writing – review and editing. Rishi Gupta: Writing – review and editing, Supervision, Project administration, Conceptualization, Methodology.

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Declarations

Availability of data and materials

The air quality dataset used in this study is publicly available from the Central Pollution Control Board (CPCB), Government of India, through the Central Control Room (CCR) air quality monitoring portal: https://airquality.cpcb.gov.in/ccr/. The dataset includes measurements from CPCB monitoring stations in Jaipur obtained from the Central Control Room (CCR) portal.

Competing interests

The authors declare no competing interests.

Ethical approval

Not applicable.

Consent to participate and consent to publish

Not applicable.

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